

AYAAZ YASIN

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Research Interests

Liquid–vapor phase change and interfacial transport, computational fluid dynamics, molecular dynamics, kinetic theory, multiscale modeling, contact-line and thin film dynamics, with applications in cryogenic propellant management, microgravity thermal systems, and heat transfer technologies.

Education

PhD in Mechanical Engineering	2024 – now
Advanced to PhD Candidacy	Jun 2026
University of Cincinnati, Cincinnati, OH	
MS in Aerospace Engineering	2024
University of Cincinnati, Cincinnati, OH	
Thesis: Computational modeling of evaporation without tuning coefficients	
BS in Mechanical Engineering Technology	2022
Minor in Mathematics	
University of Cincinnati, Cincinnati, OH	
Senior project: Aerodynamic optimization of a solar car	

Peer-Reviewed Journal Publications

4. An experimental study of interfacial dynamics control using temperature-sensitive surfactants
A. Sarchami, S. Pakanati, A. Yasin, M. Jog, and K. Bellur
Langmuir, 42 (20), doi: 10.1021/acs.langmuir.5c06748 2026
3. A multiscale CFD model of evaporating Hydrogen menisci: incorporating subgrid thin-film dynamics and in situ accommodation coefficients
A. Yasin*, S. Pakanati*, and K. Bellur
Fuels, 7 (1), doi: 10.3390/fuels7010003 2026
2. Computational modeling of evaporation without tuning coefficients
A. Yasin and K. Bellur
Applied Thermal Engineering, 276, doi: 10.1016/j.applthermaleng.2025.126807 2025
1. An investigation of phase change induced Marangoni-dominated flow patterns using the Constrained Vapor Bubble data from ISS experiments
U. Chakrabarti*, A. Yasin*, K. Bellur, and J. Allen
Frontiers in Space Technologies, 4, doi: 10.3389/frspt.2023.1263496 2023

*shows equal contribution

Invited Talks

1. Liquid-vapor phase change in aerospace applications
Dept of Aerospace Engineering and Engineering Mechanics, University of Cincinnati Apr 2025

Conference Presentations

10. Modeling multiscale oscillations in thin liquid films
A. Yasin, U. Chakrabarti, and K. Bellur
11th ASTFE Thermal and Fluids Engineering Conference, Tempe, AZ Mar 2026

9. Stability and contact line dynamics of evaporating thin liquid films
A. Sarchami, S. Pakanati, T. Enam, A. Yasin, and K. Bellur
11th ASTFE Thermal and Fluids Engineering Conference, Tempe, AZ Mar 2026
8. Multiscale oscillations in thin liquid films
A. Yasin, U. Chakrabarti, and K. Bellur
ASME International Mechanical Engineering Congress & Exposition, Memphis, TN Nov 2025
7. Exploring two-dimensional flows in evaporating thin films: a step towards a dynamic model
A. Yasin and K. Bellur
10th ASTFE Thermal and Fluids Engineering Conference, Washington, DC Mar 2025
6. Modeling of evaporation in cryogenic fuels without tuning coefficients
A. Yasin and K. Bellur
35th NASA Thermal and Fluids Analysis Workshop Aug 2024
5. Modeling evaporation without tuning coefficients
A. Yasin and K. Bellur
51st Midwestern University Fluid Mechanics Retreat, Rochester, IN Apr 2024
4. A numerical study of coefficient-free kinetic evaporation modeling in liquid Hydrogen
A. Yasin and K. Bellur
76th APS Division of Fluid Dynamics Annual Meeting, Washington, DC Nov 2023
3. An investigation of Marangoni induced flow in Constrained Vapor Bubble ISS experiments
A. Yasin, U. Chakrabarti, K. Bellur, and J. Allen
50th Midwestern University Fluid Mechanics Retreat, Rochester, IN Apr 2023
2. A CFD model of evaporation in liquid Hydrogen without the need for tuning coefficients
A. Yasin and K. Bellur
75th APS Division of Fluid Dynamics Annual Meeting, Indianapolis, IN Nov 2022
1. A solution to the 2022 AUVSI Student Unmanned Aerial Systems competition
A. Yasin, R. Gilligan, D. Heitmeyer, and K. Cohen
AIAA Region III Student Conference, Purdue University, West Lafayette, IN Mar 2022

Honors and Awards

Lester Nenninger Award, UC College of Engineering and Applied Science	2026
University Research Council Graduate Student Stipend Award, UC Office of Research	2026
Research Fellowship, UC Graduate College and Graduate Student Government	2026
Prof. Kirti Ghia Endowed Graduate Student Scholarship, UC Dept of Mechanical Engineering	2025
Excellence in Teaching Award (Honorable Mention), UC Graduate College	2024
Travel Grant, American Physical Society, Division of Fluid Dynamics	2023
Graduate Assistant Scholarship, UC Dept of Engineering and Computing Education	2023 – 2024
P&G Simulation Center Student Support Scholarship	2022
Graduate Incentive Scholarship, UC College of Engineering and Applied Science	2022 – 2026
Several conference travel awards, UC Graduate College	2022 – 2025
Undergraduate Research Fellowship, UC Office of Research	2022
Outstanding Senior Award, UC College of Engineering and Applied Science	2022

Mentoring and Supervision

Mentored undergraduate students in the UC Lab for Interfacial Dynamics.	2024 – now
Supervised a team of six undergraduate and two graduate teaching assistants.	2023 – 2024
Served as a mentor for the First-Year Engineering program.	2023 – 2024

Academic Service

Served on a University-wide committee advising graduate programs on student handbooks. 2026
Served as mechanical chair of the department's graduate student association. 2026 – 2027
Reviewer: Physics of Fluids, ACS Nano Letters, AIAA Journal of Thermophysics and Heat Transfer, ASME Journal of Heat and Mass Transfer, and Communications in Theoretical Physics.

Teaching Experience

As instructor of record

MET4076 Applied Computational Methods (Lecture & Lab) Spring 2025
ENED1120 Foundations of Engineering Design Thinking II Spring 2024
ENED1100 Foundations of Engineering Design Thinking I Spring 2023, Fall 2023

As teaching assistant

MET5036L Thermal Environmental Systems & Heat Transfer Lab Spring 2026
ENED1120 Foundations of Engineering Design Thinking II Spring 2022
ENED1100 Foundations of Engineering Design Thinking I Fall 2020, Fall 2021

Professional Experience

Research Assistant, Lab for Interfacial Dynamics (advised by Dr. Kishan Bellur) 2022 – now
Dept of Mechanical, Materials, and Industrial Engineering, University of Cincinnati
Instructor, Mechanical Engineering Spring 2025
University of Cincinnati
Instructor, Engineering and Computing Education 2023 – 2024
University of Cincinnati
Research Assistant, Proctor & Gamble Digital Accelerator Fall 2022
University of Cincinnati
Student Worker, Ohio Innocence Project Summer 2022
University of Cincinnati
Product Development Engineering Co-op Spring & Summer 2021
GMi Companies, Lebanon, OH
Manufacturing Engineering Co-op Spring & Fall 2019
Regal Rexnord, Florence, KY

Skills

Programming

Matlab, C, C++, Python, VBA, Bash, Git, L^AT_EX.

Modeling

Ansys Fluent, COMSOL Multiphysics, FEniCSx, LabView, LAMMPS, OpenFOAM, Simcenter 3D, SolidWorks, Star CCM+.